

BRADEN SCOTT *Mechatronics Engineer*

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EDUCATION

BASc, Mechanical Engineering — Mechatronics,
University of British Columbia | 2018-2024

INTERESTS

Tech | Investing | Sports | Travelling | Weightlifting

WORK EXPERIENCE

IFD Technologies

Hardware Engineer

Sep 2025 – Present

- Independently researched and designed a custom split-core PCB Rogowski coil sensor, delivering a high-bandwidth, high-accuracy, low-cost, current measurement solution for utility grid monitoring applications
- Designed and commissioned a high-current test infrastructure from the ground up, sourcing primary injection test equipment, developing all test plans, and executing full validation testing and troubleshooting of current transformer assemblies up to 1000A
- Designed a signal conditioning circuit to capture high-frequency transient events at low ADC sample rates, reducing system cost and complexity without sacrificing measurement capability
- Executed hardware bring-up, PCB rework, and validation testing across analog and RF systems using oscilloscopes, multimeters, VNA, and spectrum analyzers, with hands-on fault diagnosis and troubleshooting throughout

Mechanical Engineer

Dec 2024 – Sep 2025

- Designed mechanical components in CAD for deployment inside oil-filled power transformers, ensuring all parts met harsh thermal, chemical, and vibration environmental requirements
- Led on-site certification inspections at customer manufacturing facilities to qualify installation of company products, ensuring compliance with process and safety requirements

Product Engineer Intern

May 2021 – May 2022

- Designed and validated a patented bidirectional pressure relief valve that resolved an active nonconformance across 20,000+ customer field units, eliminating a systemic product defect at scale

Tesla , *Manufacturing Engineer Intern, BIW Pilot Team*

Jan 2023 – May 2023

- Designed manufacturing fixtures to NAAMS standards for pilot and beta build assembly of new vehicle programs, supporting production readiness ahead of full-scale launch
- Programmed and trained robotic arm cells for autonomous assembly operations, coordinating integration with controls and electronics teams
- Operated Haas CNC mill and programmed toolpaths in Fusion 360 to produce custom precision parts for manufacturing and assembly applications

Cellula Robotics LTD. , *Autonomous Underwater Vehicle Engineer Co-op*

May 2023 – Sep 2023

- Planned and conducted field deployment testing of autonomous underwater vehicles, executing pre-deployment verification of pressure-rated hardware, vehicle communications, and subsea systems in open-water environments

PROJECTS

NoBreak - Ankle Brace Startup

- Led design and validation of a flexible ankle brace achieving 150%+ improved mobility over rigid braces, including custom test fixture fabrication and FEA structural analysis in Ansys
- Conducted customer discovery interviews and market validation to develop the product value proposition and business case, applying lean startup methodology within a structured new venture development program

Study Robot Crocodile

- Designed a companion robot to improve user focus by simulating a study buddy presence, integrating motion, expression, and real-time user monitoring into a standalone embedded system
- Developed firmware on an MSP430 microcontroller to drive stepper and servo motors and animate LED displays to express emotion-based responses

SKILLS

Electrical/Software: PCB Design, Analog & RF Electronics, Signal Conditioning, ADC Signal Processing, Python, MATLAB, Embedded Firmware, Oscilloscope & DMM, VNA & Spectrum Analyzer, Soldering & PCB Rework, High-Current Testing

Mechanical: SolidWorks, Fusion 360, CATIA V5/V6, OnShape, GD&T, DFM/DFA, FMEA, Jigs & Fixtures

Manufacturing: CNC Machining (Haas), Robot Programming, FDM/SLA 3D Printing, Waterjet Cutting, PLC Programming, ISO 9001, LEAN